

Chapter 37

Pixel-Processing Accelerator (PPA)

37.1 Overview

ESP32-P4 includes a pixel-processing accelerator (PPA) designed for 2D image display. This module realizes hardware-level acceleration of image algorithms and implements functions such as image rotation, scaling, mirroring, and blending.

PPA consists of two functional modules: scaling-rotation-mirroring (SRM) and image blending (BLEND). The SRM module implements image rotation, scaling, and XY-axis mirroring functions, while the BLEND module achieves the blending of two layers of the same size based on the Alpha channel, i.e., transparency.

PPA processes images in units of pixel block. These pixel blocks are acquired from memory through 2D-DMA. The connection between PPA and 2D-DMA is shown in Figure 371-1.

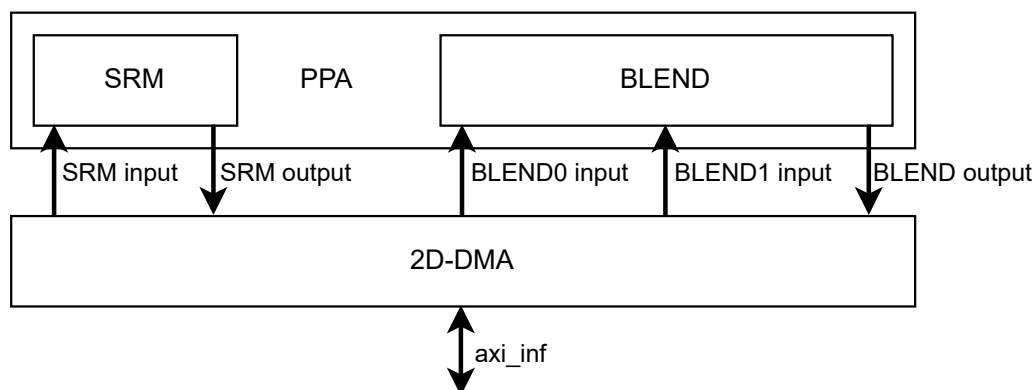


Figure 371-1. PPA 2D-DMA Connection

37.2 Terminology

To better illustrate the functions of the PPA module, the following terms are used in this chapter:

- image** A complete image stored in the system memory
- image block** A portion cropped from an image at a certain size, with the maximum size equivalent to the entire image. i.e., hb, vb size under 2D-DMA 2D-MOD0.
- pixel block** A portion cropped from an image block. SRM reads and writes data in pixel blocks.

37.3 Features

- Image rotation, scaling, and mirroring by SRM:

- Input formats: ARGB8888, RGB888, RGB565, YUV422, YUV420, GRAY
- Output formats: ARGB8888, RGB888, RGB565, YUV422, YUV420, GRAY
- Counterclockwise rotation angles: 0°, 90°, 180°, 270°
- Horizontal and vertical scaling with scaling factors of 8-bit integer part and 4-bit fractional part
- Horizontal and vertical mirroring
- Blending two layers of the same size and filling images with specific pixels by BLEND:
 - Foreground input formats: ARGB8888, RGB888, RGB565, L4, L8, A4, A8
 - Background input formats: ARGB8888, RGB888, RGB565, YUV422, YUV420, GRAY, L4, L8
 - Output formats: ARGB8888, RGB888, RGB565, YUV422, YUV420, GRAY
 - Layer blending based on the Alpha channel. If layers lack an Alpha channel, it can be provided through register configuration.
 - Special color filtering by setting color-key ranges of foreground and background layers

37.4 Architectural Overview

As shown in figure 37.4-1, there are two independent functional modules in PPA: SRM and BLEND.

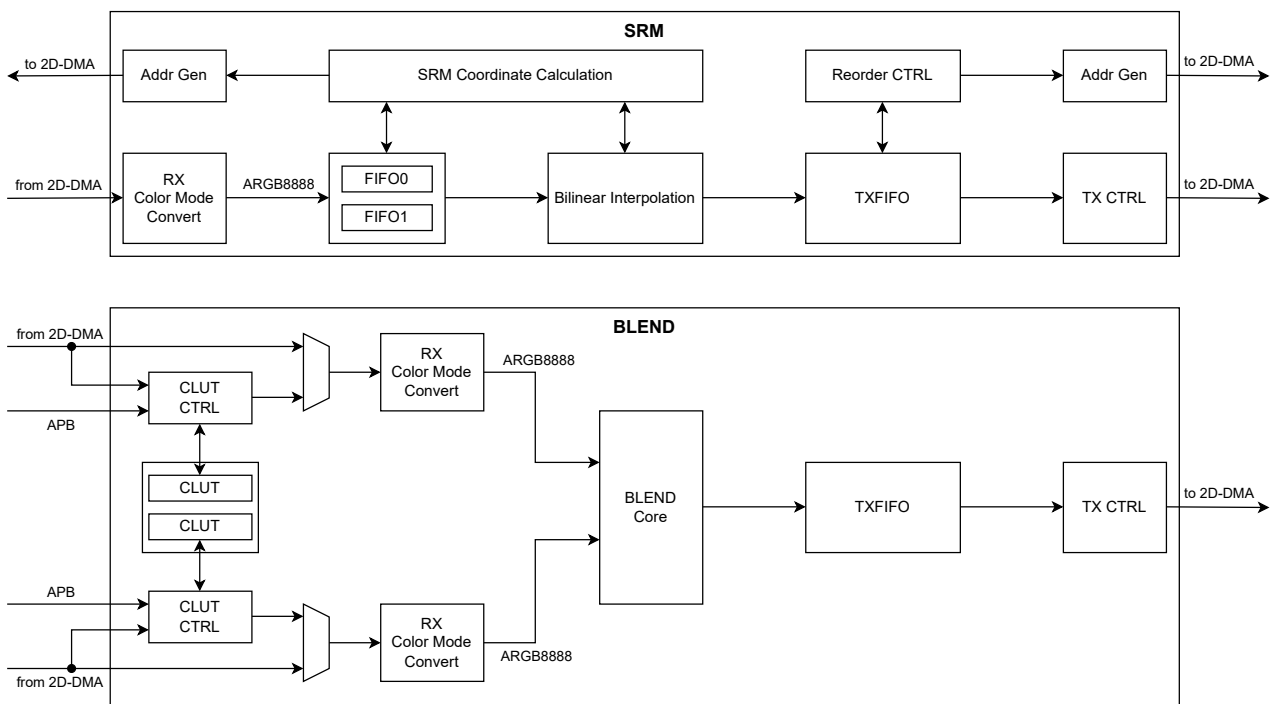


Figure 37.4-1. PPA Architecture

For SRM, SRM Coordinate Calculation scans coordinates of the output pixel blocks and calculates coordinates of the corresponding input pixel blocks. Addr Gen calculates the address according to the input pixel block coordinates and sends them to 2D-DMA. After that, the PPA receives the pixel block data from 2D-DMA, converts it into ARGB8888 format according to the register configuration, and stores it into the RX FIFO (a ping-pong FIFO composed of two SRAMs, each of which can store the whole pixel block). When a complete

pixel block is cached, Bilinear Interpolation reads the input pixels from RX FIFO, interpolates them to get the output pixels, and outputs them to TX FIFO. Reorder CTRL then calculates the output pixel block coordinates and address according to the register configuration and sends them to 2D-DMA. At the same time, Reorder CTRL takes out the corresponding pixels from TX FIFO according to the output pixel block coordinates and sends them to 2D-DMA, which saves them to the corresponding address.

For BLEND, PPA also receives data from 2D-DMA and converts it to ARGB8888 format based on the input data type. Yet the converted pixels will be processed directly by the BLEND Core algorithm module, enter the TX FIFO, and ultimately output to the system memory through 2D-DMA. Unlike SRM, BLEND does not need to calculate the address itself as it is completely controlled by 2D-DMA through a linked list.

37.5 Functional Description

37.5.1 PPA Color Space

PPA supports both true color and pseudo color. True color refers to a color representation in which each pixel in a color image consists of three primary color components: R (red), G (green), and B (blue). Each of these components directly determines the intensity of the corresponding primary color on the display device. The true color formats supported by PPA include ARGB8888, RGB888, and RGB565.

Pseudo color refers to a color representation in which the color of each pixel in a color image is not directly determined by the numerical values of the primary color components. Instead, the pixel value serves as an entry into a Color Look-Up Table (CLUT) to retrieve the corresponding A, R, G, and B values. PPA supports CLUT entry addresses with 8-bit or 4-bit width.

In addition, SRM supports the YUV422, YUV420, and GRAY data formats. The storage order of supported color formats in memory is shown in Table 37.5-1. The input YUV422 storage order can be set using the SRM or BLEND YUV422_BYTE_ORDER register; the output, however, is always stored with YUV422_BYTE_ORDER = 1.

Table 37.5-1. Pixel Storage Order

Color Format	Start Addr + 3	Start Addr + 2	Start Addr + 1	Start Addr + 0
ARGB8888	A0[7:0]	R0[7:0]	G0[7:0]	B0[7:0]
RGB888	B1[7:0]	R0[7:0]	G0[7:0]	B0[7:0]
	G2[7:0]	B2[7:0]	R1[7:0]	G1[7:0]
	R3[7:0]	G3[7:0]	B3[7:0]	R2[7:0]
RGB565	R1[4:0]G1[5:3]	G1[2:0]B1[4:0]	R0[4:0]G0[5:3]	G0[2:0]B0[4:0]
L8	L3[7:0]	L2[7:0]	L1[7:0]	L0[7:0]
L4	L7[3:0]L6[3:0]	L5[3:0]L4[3:0]	L3[3:0]L2[3:0]	L1[3:0]L0[3:0]
A8	A3[7:0]	A2[7:0]	A1[7:0]	A0[7:0]
A4	A7[3:0]A6[3:0]	A5[3:0]A4[3:0]	A3[3:0]A2[3:0]	A1[3:0]A0[3:0]
YUV422 YUV422_BYTE_ORDER=0	Y1[7:0]	V1[7:0]	Y0[7:0]	U0[7:0]

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Table 37.5-1 – ...Continued from the previous page

Color Format	Start Addr + 3	Start Addr + 2	Start Addr + 1	Start Addr + 0
YUV422 YUV422_BYTE_ORDER=1	Y1[7:0]	U1[7:0]	Y0[7:0]	V0[7:0]
YUV422 YUV422_BYTE_ORDER=2	v1[7:0]	y1[7:0]	U0[7:0]	Y0[7:0]
YUV422 YUV422_BYTE_ORDER=3	U1[7:0]	Y1[7:0]	V0[7:0]	Y0[7:0]
YUV420 Odd rows such as 1, 3, 5...	U2[7:0]	Y1[7:0]	Y0[7:0]	U0[7:0]
	Y4[7:0]	U4[7:0]	Y3[7:0]	Y2[7:0]
	Y7[7:0]	Y6[7:0]	U6[7:0]	Y5[7:0]
YUV420 Even rows such as 2, 4, 6...	V2[7:0]	Y1[7:0]	Y0[7:0]	V0[7:0]
	Y4[7:0]	V4[7:0]	Y3[7:0]	Y2[7:0]
	Y7[7:0]	Y6[7:0]	V6[7:0]	Y5[7:0]
GRAY	P3[7:0]	P2[7:0]	P1[7:0]	P0[7:0]

37.5.1.1 SRM Color Space

SRM supports ARGB8888, RGB888, RGB565, and YUV422, YUV420, and GRAY formats for both input and output. The input format is configured by `PPA_SRM_RX_CM`, while the output format is configured by `PPA_SRM_TX_CM`. The SRM input pixel processing steps are shown in Figure 37.5-1.

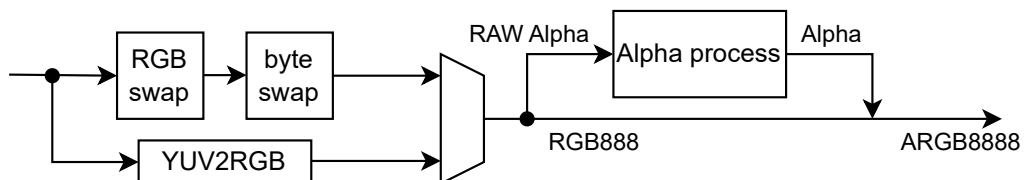


Figure 37.5-1. PPA SRM Input Pixel Processing

SRM supports RGB and byte swapping for ARGB8888, RGB888, and RGB565 input formats. The specific configurations are as follows:

- Set `PPA_SRM_RX_RGB_SWAP_EN` to 1 to reverse RGB to BGR and ARGB to BGRA.
- Set `PPA_SRM_RX_BYTE_SWAP_EN` to 1 to reverse bytes 0, 1 and 2, 3 respectively, i.e., {0, 1, 2, 3} will be reverted to {1, 0, 3, 2}. RGB888 would not be affected by this configuration as it only has three bytes per pixel.

SRM uses ARGB8888 when processing images internally. When the input lacks an Alpha channel, the default Alpha channel value for the image is set to 255. The Alpha channel can be further configured using the `PPA_SRM_FIX_ALPHA_REG` register, as shown in Figure 37.5-2. For specific configuration effects, please refer to Table 37.5-2.

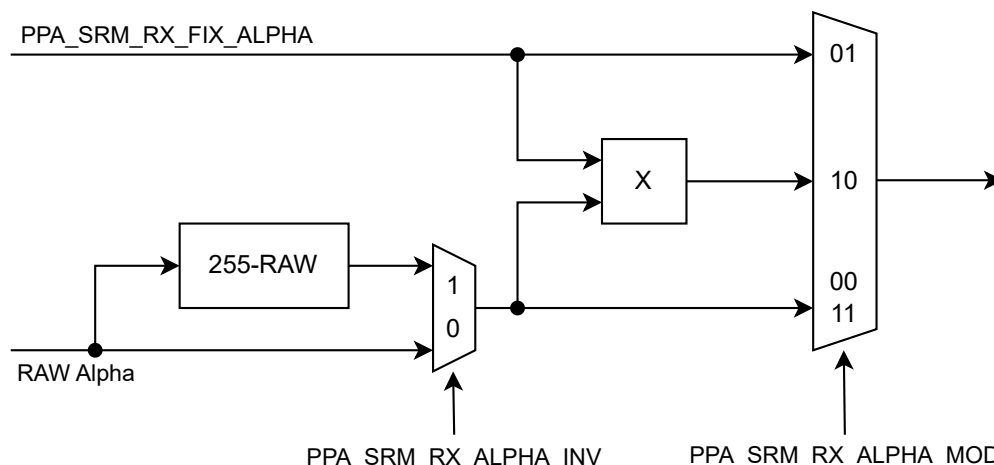


Figure 37.5-2. PPA SRM Alpha Configuration Logic

Table 37.5-2. PPA SRM Alpha Configuration

PPA_SRM_RX_ALPHA_INV	PPA_SRM_RX_ALPHA_MOD	Output
0	00/11	Original Alpha value
	01	The configuration value of PPA_SRM_RX_FIX_ALPHA
	10	The high 8 bits of the value obtained by multiplying the original Alpha value with PPA_SRM_RX_FIX_ALPHA
1	00/11	The value obtained by subtracting the original Alpha value from 255
	01	The configuration value of PPA_SRM_RX_FIX_ALPHA
	10	The high 8 bits of the value obtained by multiplying the result of subtracting the original Alpha value from 255 by PPA_SRM_RX_FIX_ALPHA

Note that when the input or output format is YUV, further configuration of YUV to RGB color conversion is required.

When the input format is YUV420, the input YUV data range needs to be configured as full-range or limit-range via the [PPA_YUV_RX_RANGE](#) field, and the protocol used for YUV to RGB conversion needs to be configured as BT601 or BT709 via the [PPA_YUV2RGB_PROTOCOLAL](#) field. The conversion formulas are as follows:

- YUV_{full} to YUV_{limit}

- $Y_{limit} = \frac{220}{256}Y_{full} + 16$

- $U_{limit} = \frac{225}{256}U_{full} + 16$

- $V_{limit} = \frac{225}{256}V_{full} + 16$

- YUV_{limit} to RGB in BT601

- $R = \frac{298}{256}Y_{limit} + \frac{409}{256}V_{limit} - \frac{56906}{256}$

- $G = \frac{298}{256}Y_{limit} - \frac{100}{256}U_{limit} - \frac{208}{256}V_{limit} + \frac{34707}{256}$

- $B = \frac{298}{256}Y_{limit} + \frac{516}{256}U_{limit} - \frac{70836}{256}$

- YUV_{limit} to RGB in BT709

$$\begin{aligned} - R &= \frac{298}{256}Y_{limit} + \frac{459}{256}V_{limit} - \frac{63367}{256} \\ - G &= \frac{298}{256}Y_{limit} - \frac{55}{256}U_{limit} - \frac{136}{256}V_{limit} + \frac{19681}{256} \\ - B &= \frac{298}{256}Y_{limit} + \frac{541}{256}U_{limit} - \frac{73918}{256} \end{aligned}$$

When the output format is YUV420, the output YUV data range needs to be configured as full-range or limit-range via the [PPA_YUV_TX_RANGE](#) field, and the protocol used for YUV to RGB conversion needs to be configured as BT601 or BT709 via the [PPA_RGB2YUV_PROTOCOL](#) field. The conversion formulas are as follows:

- YUV_{limit} to YUV_{full}

$$\begin{aligned} - Y_{full} &= \frac{298}{256}Y_{limit} + \frac{4768}{256} \\ - U_{full} &= \frac{291}{256}U_{limit} + \frac{4550}{256} \\ - V_{full} &= \frac{291}{256}V_{limit} + \frac{4550}{256} \end{aligned}$$

- RGB to YUV_{limit} in BT601

$$\begin{aligned} - Y_{limit} &= \frac{4096}{256} + \frac{66}{256}R + \frac{129}{256}G + \frac{25}{256}B \\ - U_{limit} &= \frac{32768}{256} - \frac{38}{256}R - \frac{74}{256}G + \frac{112}{256}B \\ - V_{limit} &= \frac{32768}{256} + \frac{112}{256}R - \frac{94}{256}G - \frac{18}{256}B \end{aligned}$$

- RGB to YUV_{limit} in BT709

$$\begin{aligned} - Y_{limit} &= \frac{4096}{256} + \frac{47}{256}R + \frac{157}{256}G + \frac{16}{256}B \\ - U_{limit} &= \frac{32768}{256} - \frac{26}{256}R - \frac{86}{256}G + \frac{112}{256}B \\ - V_{limit} &= \frac{32768}{256} + \frac{112}{256}R - \frac{102}{256}G - \frac{10}{256}B \end{aligned}$$

37.5.1.2 BLEND Color Space

BLEND supports ARGB8888, RGB888, RGB565, L4, L8, A4, and A8 foreground input formats, and ARGB8888, RGB888, RGB565, YUV422, YUV420, GRAY, L4, L8 background input formats. The input color format of the foreground and background layers can be configured respectively via the [PPA_BLEND0/1_RX_CM](#) field. Among the above input formats, A4 and A8 only support the foreground layer, and when the input format is L4 or A4, the size of the image block, hb, and the offset in the image, x, must be an even number. For the output format, BLEND supports ARGB8888, RGB888, and RGB565, which can be configured via the [PPA_BLEND_TX_CM](#) field.

BLEND's RGB conversion, byte conversion, and alpha channel configuration are consistent with SRM input. Additionally, when the foreground layer data format is A4 or A8, the input data will be Alpha channel values. In this case, if [PPA_BLEND_BYPASS](#) is set as 1, the foreground layer's input Alpha value replaces the background layer's Alpha channel and output. If set as 0, it will be a normal BLEND operation, and the foreground layer's RGB values will be specified by [PPA_BLEND_RGB_REG](#).

When the input data format is L4 or L8, it requires to first initialize the CLUT for BLEND. BLEND has two CLUTs with a depth of 256 and a width of 32. There are two initialization modes: FIFO and MEM.

- Write 0 to [PPA_APB_FIFO_MASK](#) to enter the FIFO mode.

In the FIFO mode, access to CLUT is achieved by reading/writing `PPA_RDWR_WORD_BLENDx_CLUT`, where 0 corresponds to the background channel CLUT and 1 corresponds to the foreground channel CLUT.

Each time this register is read/written, the corresponding read/write address is automatically incremented by 1. The read/write address can be reset by writing 1 and then 0 to `PPA_BLENDx_CLUT_MEM_RST`. The read address can be reset by writing 1 then 0 to `PPA_BLENDx_CLUT_MEM_RDADDR_RST`.

- Write 1 to `PPA_APB_FIFO_MASK` to enter the MEM mode.
In the MEM mode, CLUT can be directly accessed through addresses. The CLUT addresses in the PPA's bus address range are as shown in Table 37.5-3.

Table 37.5-3. BLEND CLUT Address

addr[11:10]	addr[9:2]	addr[1:0]
01: Select BLEND0 CLUT 10: Select BLEND1 CLUT	CLUT actual address, 0 ~ 255	00

37.5.2 PPA 2D-DMA Linked List Configuration

PPA exchanges data with the system memory via 2D-DMA. For both SRM and BLEND, it is necessary to configure the 2D-DMA linked list information for the required image block operations in one go. The 2D-DMA linked list is shown in Figure 37.5-3. For detailed information, please refer to 6 2D-DMA Controller (2D-DMA).

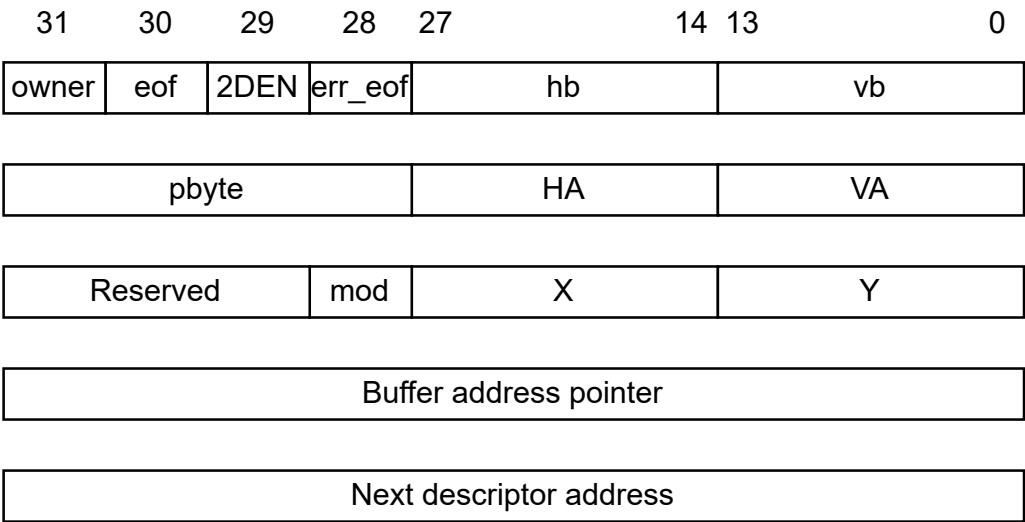


Figure 37.5-3. 2D-DMA Linked List

The corresponding linked list parameters of PPA are shown in Table 37.5-4.

Table 37.5-4. PPA Linked List Configuration

Field	Configuration
owner	1
eof	1
2DEN	1
err_eof	0
hb	Horizontal size of the image block. Must be even if the input or output data format is YUV420
vb	Vertical size of the image block. Must be even if the input or output data format is YUV420
pbyte	ARGB8888: 5 (4 byte/pixel) RGB888: 4 (3 byte/pixel) RGB565: 3 (2 byte/pixel) YUV420: 2 (1.5 byte/pixel) A8/L8: 1 (1 byte/pixel) A4/L4: 0 (0.5 byte/pixel)
HA	Horizontal size of the entire image. Must be even if the input or output data format is YUV420
VA	Vertical size of the entire image. Must be even if the input or output data format is YUV420
mod	0
X	Horizontal offset of the image block within the image (offset starts from 0). Must be even if the input or output data is YUV420
Y	Vertical offset of the image block within the image (offset starts from 0). Must be even if the input or output data is YUV420
Buffer address pointer	Starting address of the image
Next descriptor address	N/A

37.5.2.1 SRM Special Configuration

Once the 2D-DMA linked lists for input and output image blocks are configured, SRM needs to select the required pixel blocks from the input image block memory space and process them in units of pixel blocks. The processed pixel blocks are then stored in the corresponding memory space of the output image block. Throughout this process, the starting address of each pixel block needs to be controlled by SRM. To realize this functionality, write 1 to the `DMA2D_IN/OUT_DSCR_PORT_EN_CHx` field of the corresponding channel of the 2D-DMA to enable the descriptor interface (also called DSCR-PORT in 2D-DMA), allowing PPA to control

the starting address of the next pixel block in real-time.

For the data input, SRM must receive a complete pixel block each time. When the pixel block selection reaches the boundary of the image block, 2D-DMA needs to automatically pad the pixel block beyond the boundary. This can be achieved by configuring the [DMA2D_OUT_DSCR_PORT_BLK_V/H_CHx](#) field of the corresponding channel of 2D-DMA to the pixel block size. The required values depend on the image format, as shown in Table 37.5-5 below.

Table 37.5-5. Automatic Padding Block Size Configuration

PPA_SRM_BK_SIZE_SEL	Pixel Format	DMA2D_OUT_DSCR_PORT_BLK_H_CHx	DMA2D_OUT_DSCR_PORT_BLK_V_CHx
0	ARGB8888 RGB888 RGB565 GRAY	34	34
0	YUV422	36	34
0	YUV420	36	36
1	ARGB8888 RGB888 RGB565 GRAY	18	18
1	YUV422	20	18
1	YUV420	20	20

Note:

For optimal memory-access performance, use the default setting [PPA_SRM_BK_SIZE_SEL](#) = 0.

37.5.3 Scaling - Rotation - Mirroring (SRM)

37.5.3.1 Basic Functionality

SRM supports scaling, rotation, and mirroring. It first scales the image based on the origin point, then rotates it around the center point (X_{middle} , Y_{middle}), and finally mirrors it horizontally and vertically based on the center point (X_{target} , Y_{target}), as shown in Figure 37.5-4. Scaling, rotation, and mirroring parameters are all configured through registers:

- [PPA_SRM_SCAL_X_INT](#): Integer part of the horizontal scaling factor
- [PPA_SRM_SCAL_X_FRAG](#): Fractional part of the horizontal scaling factor
- [PPA_SRM_SCAL_Y_INT](#): Integer part of the vertical scaling factor
- [PPA_SRM_SCAL_Y_FRAG](#): Fractional part of the vertical scaling factor
- [PPA_SRM_ROTATE_ANGLE](#): Counterclockwise rotation angle
- [PPA_SRM_MIRROR_X](#): Enable horizontal mirroring
- [PPA_SRM_MIRROR_Y](#): Enable vertical mirroring

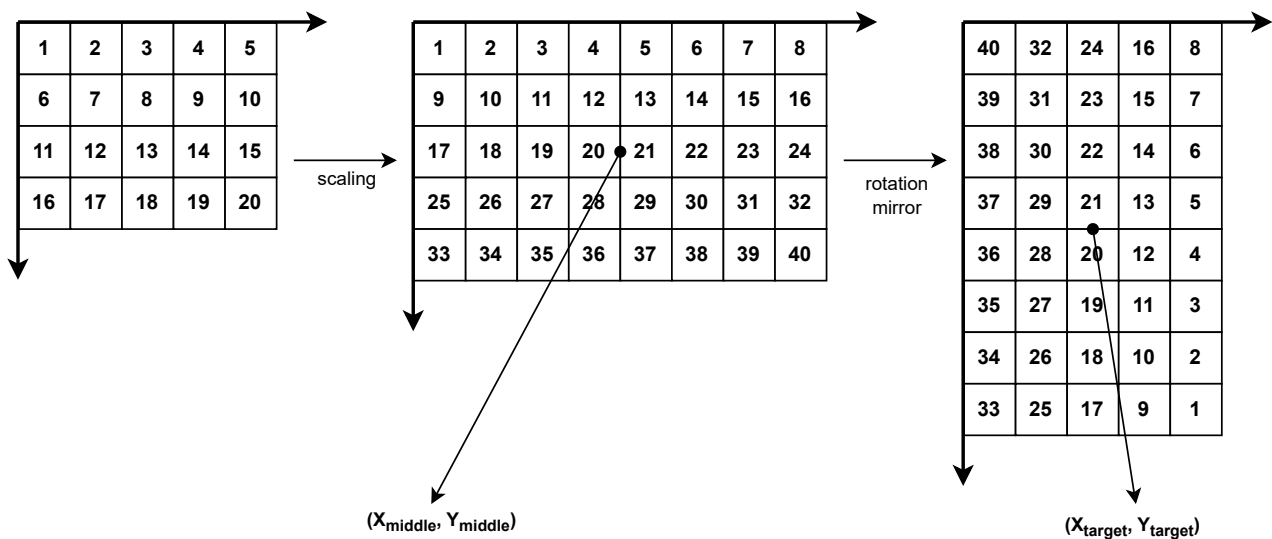


Figure 37.5-4. SRM Procedure

37.5.3.2 Pixel Block Rearrangement

When the output pixel block is small, initiating a 2D-DMA transfer for each pixel block would result in low bus bandwidth utilization. To improve bus bandwidth utilization, SRM can cache and concatenate output pixel blocks into a larger pixel block before outputting them all at once. This feature is enabled by default and can be disabled by setting `PPA_SRM_MACRO_BK_RO_BYPASS` to 1.

Once pixel block rearrangement is enabled, under certain conditions, the output pixel blocks will be automatically rearranged and concatenated, as specified in Table 37.5-6.

- For the output pixel block in ARGB8888/RGB888 format, if its vertical height is less than or equal to 32, and the horizontal width is less than or equal to 16, pixel block rearrangement will be automatically enabled.
- For the output pixel block in RGB565 format, if its vertical height is less than or equal to 32, and the horizontal width is less than or equal to 32, pixel block rearrangement will be automatically enabled.
- For the output pixel block in YUV420 format, if its vertical height is less than or equal to 32, and horizontal width is less than or equal to 32, pixel block rearrangement will be automatically enabled.

Table 37.5-6. Number of Pixel Block Rearrangements in SRM Output

ARGB8888/RGB888		RGB565		YUV420	
Horizontal Size	Rearrangement Number	Horizontal Size	Rearrangement Number	Horizontal Size (Even)	Rearrangement Number
1	32	1	64	2	32
2	16	2	32	4	16
3	10	3	20	6	10
4	8	4	16	8	8
5	6	5	12	10	6
6	5	6	10	12	5
7	4	7	9	14	4

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ARGB8888/RGB888		RGB565		YUV420	
Horizontal Size	Rearrangement Number	Horizontal Size	Rearrangement Number	Horizontal Size (Even)	Rearrangement Number
8	4	8	8	16	4
9	3	9	7	18	3
10	3	10	6	20-32	2
11	2	11	5		
12	2	12	5		
13	2	13	4		
14	2	14	4		
15	2	15	4		
16	2	16	4		
		17	3		
		18	3		
		19-32	2		

37.5.4 Layer Blending (BLEND)

37.5.4.1 Basic Functionality

BLEND combines two image blocks of the same size using the Alpha channel and then outputs the result. The blending formula is as follows, where A_b is the Alpha channel of the background layer, A_f is the Alpha channel of the foreground layer, C_b corresponds to the R, G, and B components of the background layer, and C_f corresponds to the R, G, B components of the foreground layer:

- $A_{out} = A_b + A_f - A_b * A_f$
- $C_{out} = (C_b * A_b * (1 - A_f) + C_f * A_f) / (A_b + A_f - A_b * A_f)$

Additionally, BLEND supports color-keying based on pixel color. By configuring color ranges via [PPA_CK_FG/BG_HIGH/LOW_REG](#), BLEND allows for image segmentation into four distinct regions, each subjected to different operations:

- When the foreground pixel falls within the color-key range and the background pixel falls outside the color-key range, output the background pixel.
- When the background pixel falls within the color-key range and the foreground pixel falls outside the color-key range, the output depends on the value of [PPA_COLORKEY_FG_BG_REVERSE](#). If 0, output the background pixel; if 1, output the foreground pixel.
- When both the background and foreground pixels fall within the color-key range, the output is set to the color defined by [PPA_COLORKEY_DEFAULT_R/G/B](#).
- When neither the background nor foreground pixels fall within the color-key range, output follows the normal Alpha Blending process.

When [PPA_BLEND_BYPASS](#) is set to 1, BLEND directly outputs the background layer data. If the foreground input data type is A4/A8 at this time, the foreground input data will replace the background input data's Alpha channel and then be output.

37.5.4.2 Filled Image Output

BLEND also supports filled image output. When this feature is enabled, only the output side of BLEND works, continuously outputting the filled image. The size and data of the filled image can be configured via [PPA_BLEND_TX_SIZE_REG](#) and [PPA_BLEND_TX_FIX_PIXEL](#).

37.6 Interrupts

ESP32-P4's PPA can generate the following interrupt signal that will be sent to the [Interrupt Matrix](#).

- PPA_INTR

The following interrupt sources can generate PPA_INTR interrupt signal:

- PPA_SRM_PARAM_CFG_ERR_INT: Triggered when invalid parameters are configured in the SRM mode. You can acquire the specific error type through [PPA_SRM_PARAM_ERR_ST_REG](#), as shown in Table 37.6-1.
- PPA_BLEND_PARAM_CFG_ERR_INT: Triggered when invalid parameters are configured in the BLEND mode. You can acquire the specific error type through [PPA_BLEND_ST_REG](#), as shown in Table 37.6-2.
- PPA_SRM_EOF_INT: Triggered when SRM workflow reaches EOF.
- PPA_BLEND_EOF_INT: Triggered when BLEND workflow reaches EOF.

Table 37.6-1. SRM Parameter Error Types

Error Type	Details
PPA_TX_DSCR_VB_ERR_ST	The sum of the vertical size of the output image block from SRM and the vertical offset Y specified in the output descriptor exceeds the total vertical size of the image configured in the output descriptor (VA)
PPA_TX_DSCR_HB_ERR_ST	The sum of the horizontal size of the output image block from SRM and the vertical offset X specified in the output descriptor exceeds the total horizontal size of the image configured in the output descriptor (HA)
PPA_Y_RX_SCAL_EQUAL_O_ERR_ST	The vertical scaling factor for the input image block in SRM is set to 0
PPA_RX_DSCR_VB_ERR_ST	The sum of the vertical size VB and the vertical offset Y in the SRM input descriptor exceeds the overall vertical size VA in the output descriptor
PPA_YDST_LEN_TOO_SAMLL_ERR_ST	The vertical size of the image block after SRM scaling is 0. For example, if the vertical size of the image block is 14 and the vertical scaling factor is 1/16 without rotation, the scaled vertical size of the image block becomes 0
PPA_YDST_LEN_TOO_LARGE_ERR_ST	The vertical size of the image block after SRM scaling exceeds 8191
PPA_X_RX_SCAL_EQUAL_O_ERR_ST	The horizontal scaling factor for the input image block in SRM is 0

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Table 37.6-1 – ...Continued from the previous page

Error Type	Details
PPA_RX_DSCR_HB_ERR_ST	The sum of the horizontal size HB of the input image block configured in the SRM input descriptor and the horizontal offset X exceeds the entire image's horizontal size HA configured in the output descriptor
PPA_XDST_LEN_TOO_SAMLL_ERR_ST	The horizontal size of the image block after SRM scaling is 0
PPA_XDST_LEN_TOO_LARGE_ERR_ST	The horizontal size of the image block after SRM scaling exceeds 8191
PPA_X_YUV420_RX_SCALE_ERR_ST	The input descriptor parameters HA/hb/X happen to be odd numbers when the SRM input format is YUV420
PPA_Y_YUV420_RX_SCALE_ERR_ST	The input descriptor parameters VA/vb/Y happen to be odd numbers when the SRM input format is YUV420
PPA_X_YUV420_TX_SCALE_ERR_ST	The output descriptor parameters HA/hb/X happen to be odd numbers when the SRM output format is YUV420
PPA_Y_YUV420_TX_SCALE_ERR_ST	The output descriptor parameters VA/vb/Y happen to be odd numbers when the SRM output format is YUV420

Table 37.6-2. BLEND Parameter Error Types

Error Type	Details
PPA_BLEND_SIZE_DIFF_ST	The BLEND foreground and background sizes do not match
PPA_BLEND_YUV_X_SCALE_ERR_ST	The BLEND background image format is YUV422 or YUV420, the horizontal size is an odd number
PPA_BLEND_YUV_Y_SCALE_ERR_ST	The BLEND background image format is YUV420, the vertical size is an odd number

37.7 Programming Procedures

37.7.1 PPA Clock Reset Configuration

1. Enable the clock of the PPA and 2D-DMA modules:
 - Write 1 to [HP_SYS_CLKRST_PPA_SYS_CLK_EN](#) to enable the PPA clock
 - Write 1 to [HP_SYS_CLKRST_DMA2D_SYS_CLK_EN](#) to enable the 2D-DMA clock
2. Reset the PPA and 2D-DMA modules:
 - Write 1 and then 0 to [HP_SYS_CLKRST_RST_EN_PPA](#) reset PPA
 - Write 1 and then 0 to [HP_SYS_CLKRST_RST_EN_DMA2D](#) reset 2D-DMA

37.7.2 SRM Configuration

1. Refer to Section [37.7.1](#) to enable and reset PPA and 2D-DMA.
2. Configure the inlink and outlink of 2D-DMA. For detailed parameters, please refer to Table [37.5-4](#).

3. Select the input and output channels of 2D-DMA:
 - Write 0x1 to [DMA2D_OUT_PERI_SEL_CHx](#) to allocate one 2D-DMA output channel to the PPA SRM input
 - Write 0x1 to [DMA2D_IN_PERI_SEL_CHx](#) to allocate one 2D-DMA input channel to the PPA SRM output
4. Enable descriptor interface for 2D-DMA channels used by PPA:
 - Write 1 to [DMA2D_OUT_DSCR_PORT_EN_CHx](#) to enable descriptor interface
 - Write 1 to [DMA2D_IN_DSCR_PORT_EN_CHx](#) to enable descriptor interface
5. Configure the descriptor interface completion parameters of 2D-DMA:
 - Configure the horizontal completion parameters via [DMA2D_OUT_DSCR_PORT_BLK_H_CHx](#). See Table 37.5-5 for details
 - Configure the vertical completion parameters via [DMA2D_OUT_DSCR_PORT_BLK_V_CHx](#). See Table 37.5-5 for details
6. Enable the input and output channels of 2D-DMA:
 - Write 1 to [DMA2D_OUTLINK_START_CHx](#) to enable the output channel of 2D-DMA
 - Write 1 to [DMA2D_INLINK_START_CHx](#) to enable the input channel of 2D-DMA
7. Configure SRM parameters:
 - Configure the size of image blocks to be processed via [PPA_SRM_BK_SIZE_SEL](#)
 - Configure SRM input data format via [PPA_SRM_RX_CM](#)
 - Configure SRM output data format via [PPA_SRM_TX_CM](#)
 - Configure the integer part of the horizontal scaling factor via [PPA_SRM_SCAL_X_INT](#)
 - Configure the fractional part of the horizontal scaling factor via [PPA_SRM_SCAL_X_FRAG](#)
 - Configure the integer part of the vertical scaling factor via [PPA_SRM_SCAL_Y_INT](#)
 - Configure the fractional part of the vertical scaling factor via [PPA_SRM_SCAL_Y_FRAG](#)
 - Configure the counterclockwise rotation angle via [PPA_SRM_ROTATE_ANGLE](#)
 - Enable horizontal mirroring via [PPA_SRM_MIRROR_X](#)
 - Enable vertical mirroring via [PPA_SRM_MIRROR_Y](#)
8. Write 1 to [PPA_SCAL_ROTATE_START](#) to start the SRM workflow.

37.7.3 BLEND CLUT Configuration

1. In the FIFO mode:
 - Write 0 to [PPA_APB_FIFO_MASK](#) to enter the FIFO mode
 - Write data to [PPA_RDWR_WORD_BLEND0_CLUT](#) to initialize the background layer's CLUT
 - Write data to [PPA_RDWR_WORD_BLEND1_CLUT](#) to initialize the foreground layer's CLUT
2. In the MEM mode:
 - Write 1 to [PPA_APB_FIFO_MASK](#) to enter the MEM mode

- Initialize the CLUT by writing data directly to the corresponding address, referring to Table [37.5-3](#)

37.7.4 BLEND Configuration

1. Refer to Section [37.7.1](#) to enable and reset PPA and 2D-DMA.
2. Configure the input and output link lists of 2D-DMA. For detailed parameters, please refer to Table [37.5-4](#).
3. Select the input and output channels of 2D-DMA:
 - Write 0x2 to [DMA2D_OUT_PERI_SEL_CHx](#) to allocate one 2D-DMA output channel to PPA BLEND for background layer input
 - Write 0x3 to [DMA2D_OUT_PERI_SEL_CHx](#) to allocate one 2D-DMA output channel to PPA BLEND for foreground layer input
 - Write 0x2 to [DMA2D_IN_PERI_SEL_CHx](#) to allocate one 2D-DMA input channel to PPA BLEND for output
4. Enable the input and output channels for 2D-DMA. The x in the registers refers to the two channels allocated to BLEND foreground and background respectively:
 - Write 1 to [DMA2D_OUTLINK_START_CHx](#) to enable the output channel of 2D-DMA
 - Write 1 to [DMA2D_INLINK_START_CHx](#) to enable the input channel of 2D-DMA
5. Configure BLEND parameter:
 - Configure the data type for the background layer input of BLEND via [PPA_BLEND0_RX_CM](#). If the format is L4/L8, initialize the CLUT first as described in [37.7.3](#)
 - Configure the data type for the foreground layer input of BLEND via [PPA_BLEND1_RX_CM](#). If the format is L4/L8, initialize the CLUT first as described in [37.7.3](#)
 - Configure the output data format of BLEND via [PPA_BLEND_TX_CM](#)
 - Configure the upper and lower limits of the color-key for the background layer via [PPA_CK_BG_HIGH_REG](#) and [PPA_CK_BG_LOW_REG](#)
 - Configure the upper and lower limits of the color-key for the foreground layer via [PPA_CK_FG_HIGH_REG](#) and [PPA_CK_FG_LOW_REG](#)
 - Configure the default value and behavior for the color-key via [PPA_CK_DEFAULT_REG](#)
 - Write 1 to [PPA_BLEND_EN](#) to enable BLEND
6. Write 1 to [PPA_BLEND_TRANS_MODE_UPDATE](#) to start BLEND workflow

37.7.5 BLEND Image Filling Configuration

1. Refer to Section [37.7.1](#) to enable and reset PPA and 2D-DMA.
2. Configure the input link lists of 2D-DMA (output link lists of PPA BLEND). For detailed parameters, please refer to Table [37.5-4](#).
3. Write 0x2 to [DMA2D_IN_PERI_SEL_CHx](#) to allocate one 2D-DMA input channel to PPA BLEND output.
4. Write 1 to [DMA2D_INLINK_START_CHx](#) to enable the output channel of 2D-DMA.

5. Configure BLEND parameters:

- Configure the output data format of BLEND via [PPA_BLEND_TX_CM](#)
- Write 1 to [PPA_BLEND_FIX_PIXEL_FILL_EN](#) to enable image filling
- Configure the pixel value for image filling via [PPA_BLEND_TX_FIX_PIXEL](#)
- Configure the filled image size via [PPA_BLEND_HB](#) and [PPA_BLEND_VB](#)

6. Write 1 to [PPA_BLEND_TRANS_MODE_UPDATE](#) to start BLEND workflow.

37.7.6 Error Handling

- Refer to Section [37.7.1](#) to reset the whole PPA, and configure again.
- Or write 1 and then 0 to reset SRM and BLEND via [PPA_SCAL_ROTATE_RST](#), [PPA_BLEND_RST](#), and configure again.

Note:

- The 2D-DMA configurations mentioned in this chapter involve only key steps. For detailed procedures, please refer to [6 2D-DMA Controller \(2D-DMA\)](#).
- In this chapter, DMA2D_OUT corresponds to the PPA input direction, and DMA2D_IN corresponds to the PPA output direction.

37.8 Register Summary

The addresses in this section are relative to the Pixel-Processing Accelerator base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section [Access Types for Registers](#).

Name	Description	Address	Access
Configuration Registers			
PPA_BLEND0_CLUT_DATA_REG	CLUT SRAM data read/write register of BLEND background layer	0x0000	R/W
PPA_BLEND1_CLUT_DATA_REG	CLUT SRAM data read/write register of BLEND foreground layer	0x0004	R/W
PPA_CLUT_CONF_REG	Configures BLEND CLUT	0x000C	R/W
PPA_SRM_COLOR_MODE_REG	Configures SRM color space	0x0020	R/W
PPA_BLEND_COLOR_MODE_REG	Configures BLEND color space	0x0024	R/W
PPA_SRM_BYTE_ORDER_REG	Configures SRM byte order	0x0028	R/W
PPA_BLEND_BYTE_ORDER_REG	Configures BLEND byte order	0x002C	R/W
PPA_BLEND_TRANS_MODE_REG	Configures the BLEND mode	0x0034	varies
PPA_SRM_FIX_ALPHA_REG	Configures SRM Alpha channel	0x0038	R/W
PPA_BLEND_TX_SIZE_REG	Configures BLEND image filling size	0x003C	R/W
PPA_BLEND_FIX_ALPHA_REG	Configures BLEND Alpha override	0x0040	R/W
PPA_BLEND_RGB_REG	Configures RGB color	0x0048	R/W
PPA_BLEND_FIX_PIXEL_REG	Configures BLEND image filling pixel	0x004C	R/W
PPA_CK_FG_LOW_REG	Configures the foreground color-key lower threshold of BLEND	0x0050	R/W
PPA_CK_FG_HIGH_REG	Configures the foreground color-key higher threshold of BLEND	0x0054	R/W
PPA_CK_BG_LOW_REG	Configures the background color-key lower threshold of BLEND	0x0058	R/W
PPA_CK_BG_HIGH_REG	Configures the background color-key higher threshold of BLEND	0x005C	R/W
PPA_CK_DEFAULT_REG	Configures the default color-key value of BLEND	0x0060	R/W
PPA_SRM_SCAL_ROTATE_REG	Configures the SRM mode	0x0064	varies
PPA_SRM_MEM_PD_REG	Configures SRM memory	0x0068	R/W
PPA_REG_CONF_REG	Enables register clock	0x006C	R/W
PPA_RGB2GRAY_REG	Configures the coefficient in RGB to GRAY conversion	0x0098	R/W
Interrupt Registers			
PPA_INT_RAW_REG	Raw status interrupt	0x0010	R/ WTC/ SS
PPA_INT_ST_REG	Masked interrupt	0x0014	RO
PPA_INT_ENA_REG	Interrupt enable bits	0x0018	R/W

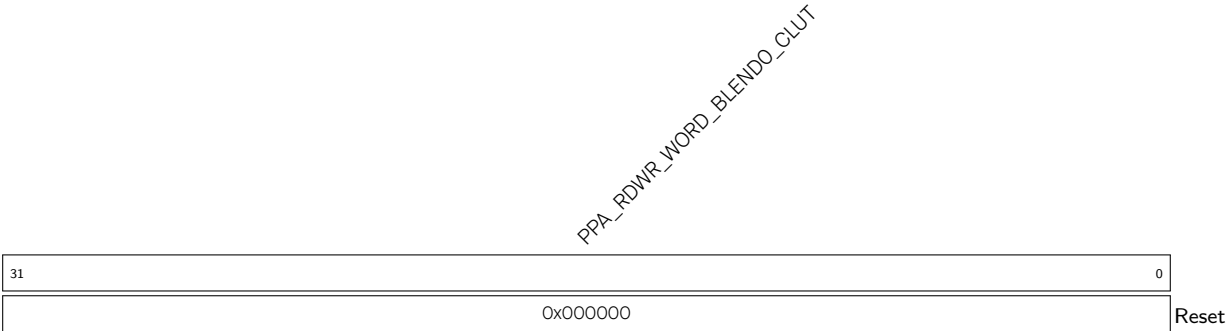
Name	Description	Address	Access
PPA_INT_CLR_REG	Interrupt clear bits	0x001C	WT
Status Registers			
PPA_CLUT_CNT_REG	BLEND CLUT write counter register	0x0070	RO
PPA_BLEND_ST_REG	BLEND status register	0x0074	RO
PPA_SRM_PARAM_ERR_ST_REG	SRM configuration error status register	0x0078	RO
PPA_SRM_STATUS_REG	SRM FSM register	0x007C	RO
Debug Register			
PPA_DEBUG_CTRL0_REG	Debug register	0x0090	R/W
PPA_DEBUG_CTRL1_REG	Debug register	0x0094	R/W
Version Register			
PPA_VERSION_REG	PPA Version register	0x0100	R/W

37.9 Registers

The addresses in this section are relative to the Pixel-Processing Accelerator base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

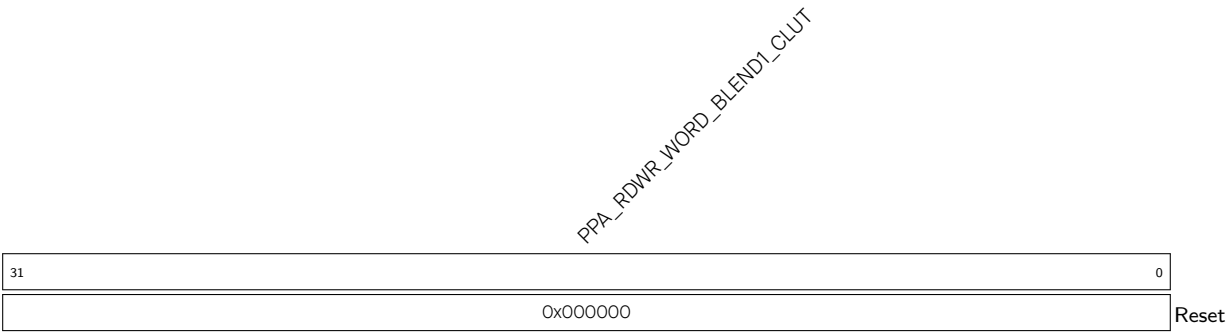
For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 37.1. PPA_BLEND0_CLUT_DATA_REG (0x0000)



PPA_RDWR_WORD_BLEND0_CLUT Configures BLEND background layer CLUT access in FIFO mode. (R/W)

Register 37.2. PPA_BLEND1_CLUT_DATA_REG (0x0004)



PPA_RDWR_WORD_BLEND1_CLUT Configures BLEND foreground layer CLUT access in FIFO mode. (R/W)

Register 37.3. PPA_CLUT_CONF_REG (0x000C)

(reserved)																PPA_BLEND0_CLUT_MEM_CLK_ENA			
																PPA_BLEND0_CLUT_MEM_FORCE_PU			
																PPA_BLEND0_CLUT_MEM_FORCE_PD			
																PPA_BLEND1_CLUT_MEM_RDADDR_RST			
																PPA_BLEND1_CLUT_MEM_RST			
																PPA_BLEND0_CLUT_MEM_RST			
																PPA_BLEND1_CLUT_MEM_RST			
																PPA_BLEND0_CLUT_MASK			
																PPA_APB_FIFO_MASK			
31								8	7	6	5	4	3	2	1	0			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

PPA_APB_FIFO_MASK Configures the access mode of BLEND CLUT1'b0: Access CLUT in FIFO mode. For details, please refer to [37.5.1.2](#).1'b1: Access CLUT in MEM mode. For details, please refer to [37.5.1.2](#).

(R/W)

PPA_BLEND0_CLUT_MEM_RST Configures whether to reset BLEND0 CLUT.

0: Release reset

1: Reset

(R/W)

PPA_BLEND1_CLUT_MEM_RST Configures whether to reset BLEND1 CLUT.

0: Release reset

1: Reset

(R/W)

PPA_BLEND0_CLUT_MEM_RDADDR_RST Configures whether to reset BLEND0 CLUT read address in FIFO mode.

0: Release reset

1: Reset

(R/W)

PPA_BLEND1_CLUT_MEM_RDADDR_RST Configures whether to reset BLEND1 CLUT read address in FIFO mode.

0: Release reset

1: Reset

(R/W)

PPA_BLEND0_CLUT_MEM_FORCE_PD Configures whether to enable the force power down of BLEND CLUT.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 37.3. PPA_CLUT_CONF_REG (0x000C)

Continued from the previous page...

PPA_BLEND0_CLUT_MEM_FORCE_PU Configures whether to enable the force power up of BLEND CLUT.

0: Disable

1: Enable

(R/W)

PPA_BLEND0_CLUT_MEM_CLK_ENA Configures whether to enable the force clock on of BLEND CLUT.

0: Disable

1: Enable

(R/W)

Register 37.4. PPA_SRM_COLOR_MODE_REG (0x0020)

(reserved)																PPA_YUV422_RX_BYTE_ORDER										PPA_RGB2YUV_PROTOCOL										PPA_YUV2RGB_PROTOCOL										PPA_YUV_TX_RANGE										PPA_YUV_RX_RANGE										PPA_SRM_TX_CM										PPA_SRM_RX_CM									
31																14																13	12	11	10	9	8	7	4				3	0																																									
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0x0																0	0	0	0	0				0				Reset																																									

PPA_SRM_RX_CM Configures the input image color format for SRM.

0: ARGB8888
1: RGB888
2: RGB565
8: YUV420
Others: Reserved
(R/W)

PPA_SRM_TX_CM Configures the output image color format for SRM.

0: ARGB8888
1: RGB888
2: RGB565
8: YUV420
Others: Reserved
(R/W)

PPA_YUV_RX_RANGE Configures the YUV range when the SRM input format is YUV.

0: limit range
1: full range
(R/W)

PPA_YUV_TX_RANGE Configures the YUV range when the SRM output format is YUV.

0: limit range
1: full range
(R/W)

PPA_YUV2RGB_PROTOCOL Configures the protocol used for the SRM input direction YUV to RGB.

0: BT601
1: BT709
(R/W)

PPA_RGB2YUV_PROTOCOL Configures the protocol used for the SRM output direction RGB to YUV.

0: BT601
1: BT709
(R/W)

Continued on the next page...

Register 37.4. PPA_SRM_COLOR_MODE_REG (0x0020)

Continued from the previous page...

PPA_YUV422_RX_BYTE_ORDER Configures the byte order when the SRM input format is YUV422.

0: YVYU

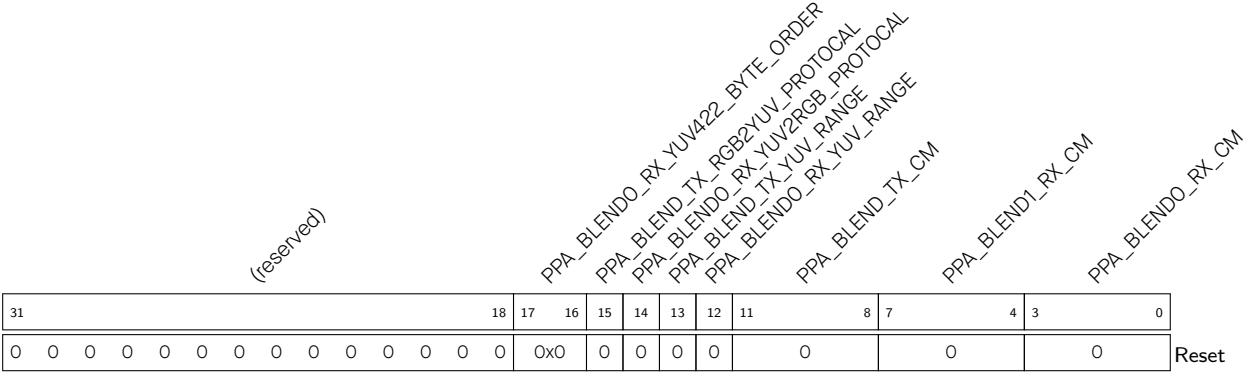
1: YUYV

2: VYUY

3: UYVY

(R/W)

Register 37.5. PPA_BLEND_COLOR_MODE_REG (0x0024)



PPA_BLEND0_RX_CM Configures the input image color format for BLEND background layer.

- 0: ARGB8888
- 1: RGB888
- 2: RGB565
- 4: L8
- 5: L4
- 8: YUV420
- 9: YUV422
- 12: GRAY
- Others: Reserved

(R/W)

PPA_BLEND1_RX_CM Configures the input image color format for BLEND foreground layer.

- 0: ARGB8888
- 1: RGB888
- 2: RGB565
- 4: L8
- 5: L4
- 6: A8
- 7: A4
- Others: Reserved

(R/W)

PPA_BLEND_TX_CM Configures the output image color format of BLEND.

- 0: ARGB8888
- 1: RGB888
- 2: RGB565
- 8: YUV420
- 9: YUV422
- 12: GRAY
- Others: Reserved

(R/W)

Continued on the next page...

Register 37.5. PPA_BLEND_COLOR_MODE_REG (0x0024)

Continued from the previous page...

PPA_BLEND_RX_YUV_RANGE Configures the YUV range when the BLEND background input is YUV.

0: Limited range

1: Full range

(R/W)

PPA_BLEND_TX_YUV_RANGE Configures the YUV range when the BLEND output is YUV.

0: Limited range

1: Full range

(R/W)

PPA_BLEND_RX_YUV2RGB_PROTOCOL Configures the YUV to RGB conversion protocol for the BLEND background input.

0: BT601

1: BT709

(R/W)

PPA_BLEND_TX_RGB2YUV_PROTOCOL Configures the YUV to RGB conversion protocol for the BLEND output.

0: BT601

1: BT709

(R/W)

PPA_BLEND_RX_YUV422_BYTE_ORDER Configures the byte order when the BLEND background input is YUV422.

0: YVYU

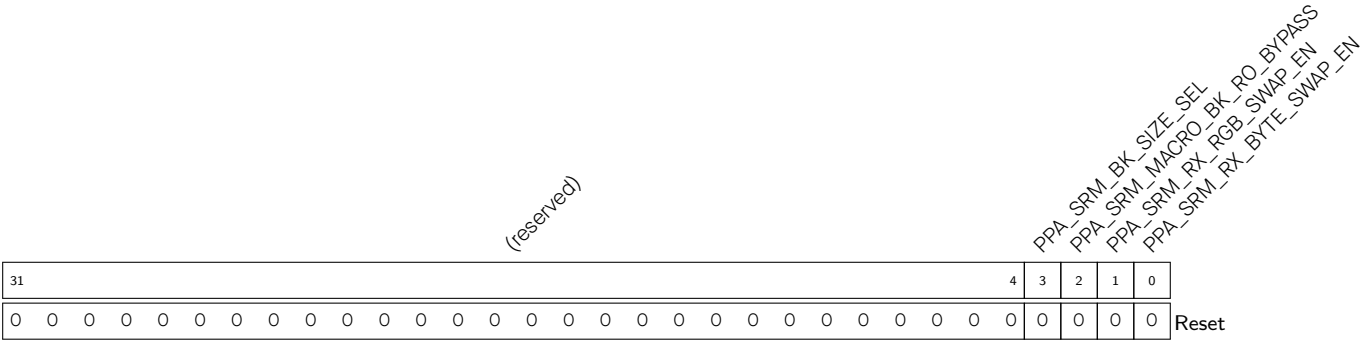
1: YUYV

2: VYUY

3: UYVY

(R/W)

Register 37.6. PPA_SRM_BYTE_ORDER_REG (0x0028)



PPA_SRM_RX_BYTE_SWAP_EN Configures whether to swap the SRM valid data input in byte.
0: Not swap
1: Swap
(R/W)

PPA_SRM_RX_RGB_SWAP_EN Configures whether to swap the order of each channel when the SRM input is RGB or ARGB.
0: Not swap
1: Swap RGB to BGR, and ARGB to BGRA
(R/W)

PPA_SRM_MACRO_BK_RO_BYPASS Configures whether to enable the pixel block order function
0: Enable and automatically calculate if the current block needs to be reordered
1: Disable
(R/W)

PPA_SRM_BK_SIZE_SEL Configures the size of SRM pixel blocks.
0: 32x32
1: 16x16
(R/W)

Register 37. PPA_BLEND_BYTE_ORDER_REG (0x002C)

Register diagram for `PPA_BLEND1_RX_RGB_SWAP_EN`. The register is 32 bits wide. Bits 31-4 are reserved. Bits 3-0 are control bits. Bit 3 is `PPA_BLEND1_RX_RGB_SWAP_EN`. Bit 2 is `PPA_BLEND1_RX_RGB_SWAP_EN`. Bit 1 is `PPA_BLEND1_RX_BYTE_SWAP_EN`. Bit 0 is `PPA_BLEND1_RX_BYTE_SWAP_EN`. The register is reset to 0.

31	4	3	2	1	0
Reset					

PPA_BLEND0_RX_BYTE_SWAP_EN	Configures whether to swap the BLEND background layer valid data input in byte.
----------------------------	---

0: Not swap

1: Swap

(R/W)

PPA_BLEND1_RX_BYTE_SWAP_EN Configures whether to swap the BLEND foreground layer valid data input in byte.

0: Not swap

1: Swap

(R/W)

PPA_BLEND0_RX_RGB_SWAP_EN Configures whether to swap the order of each channel when the BLEND background layer input is RGB or ARGB.

0: Not swap

1: Swap RGB to BGR, and ARGB to BGRA

(R/W)

PPA_BLEND1_RX_RGB_SWAP_EN Configures whether to swap the order of each channel when the BLEND foreground layer input is RGB or ARGB.

0: Not swap

1: Swap RGB to BGR, and ARGB to BGRA

(R/W)

Register 37.8. PPA_BLEND_TRANS_MODE_REG (0x0034)

[illegible]

PPA_BLEND_EN Configures whether to enable BLEND.

0: Disable

1: Enable

(R/W)

PPA_BLEND_BYPASS Configures whether to bypass BLEND and directly output the background layer.

0: Bypass

1: Not bypass

(R/W)

PPA_BLEND_FIX_PIXEL_FILL_EN Configures whether to enable filled image output.

0: Disable

1: Enable, where only BLEND TX works and the output pixel and size are configured by `PPA_BLEND_TX_SIZE_REG` and `PPA_BLEND_TX_FIX_PIXEL`.

(R/W)

PPA_BLEND_TRANS_MODE_UPDATE Write 1 to update the transfer mode, active
PPA_BLEND_FIX_PIXEL_FILL_EN, and start BLEND workflow. (WT)

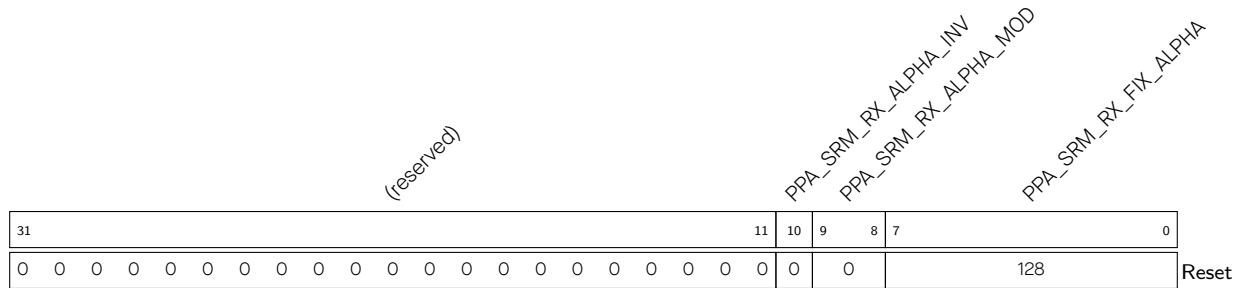
PPA_BLEND_FIX_PIXEL_FILL_EN, and start BLEND workflow. (WT)

PPA_BLEND_RST Configures whether to reset BLEND.

0: Release reset

1: Reset

(R/W)

Register 37.9. PPA_SRM_FIX_ALPHA_REG (0x0038)

PPA_SRM_RX_FIX_ALPHA Configures the Alpha channel value of SRM. (R/W)

PPA_SRM_RX_ALPHA_MOD Configures the Alpha mode of SRM input data.

1: Use the value of [PPA_SRM_RX_FIX_ALPHA](#)

2: Use the value of the higher eight bits of the original Alpha value multiplied by [PPA_SRM_RX_FIX_ALPHA](#)

Others: Use the value of the original Alpha value

(R/W)

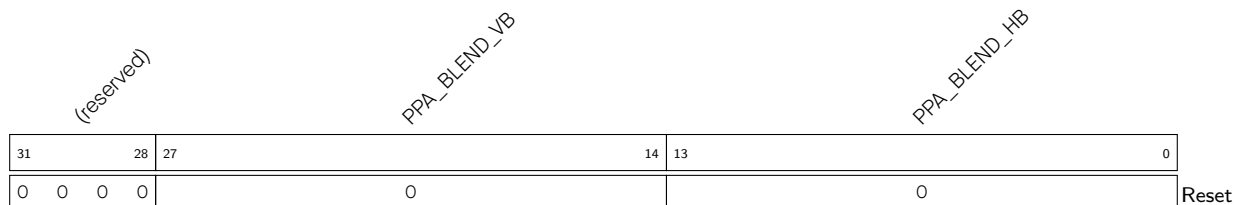
PPA_SRM_RX_ALPHA_INV Configures whether to invert the original Alpha value of SRM.

0: Not invert

1: Invert to 255 minus the original Alpha value

For data formats without Alpha channels, the original Alpha value is 255.

(R/W)

Register 37.10. PPA_BLEND_TX_SIZE_REG (0x003C)

PPA_BLEND_HB Configures the horizontal width of image block that would be filled in the image filling mode. Measurement unit: pixel. (R/W)

PPA_BLEND_VB Configures the vertical width of image block that would be filled in the image filling mode. Measurement unit: pixel. (R/W)

Register 37.11. PPA_BLEND_FIX_ALPHA_REG (0x0040)

[illegible]

PPA_BLEND0_RX_FIX_ALPHA	Configures the Alpha channel value of the BLEND background layer. (R/W)
--------------------------------	--

PPA_BLEND1_RX_FIX_ALPHA Configures the Alpha channel value of the BLEND foreground layer.
(R/W)

PPA_BLEND0_RX_ALPHA_MOD Configures the Alpha mode of BLEND background layer data.

- 1: Use the value of `PPA_BLEND0_RX_FIX_ALPHA`
- 2: Use the value of the higher eight bits of the original Alpha value multiplied by `PPA_BLEND0_RX_FIX_ALPHA`

Others: Use the value of the original Alpha value

(R/W)

PPA_BLEND1_RX_ALPHA_MOD Configures the Alpha mode of BLEND foreground layer data.

- 1: Use the value of `PPA_BLEND1_RX_FIX_ALPHA`
- 2: Use the value of the higher eight bits of the original Alpha value multiplied by `PPA_BLEND1_RX_FIX_ALPHA`

Others: Use the value of the original Alpha value

(R/W)

PPA_BLENDO_RX_ALPHA_INV Configures whether to invert the original Alpha value of the BLEND background layer data.

0: Not invert

1: Invert to 255 minus the original Alpha value

For data formats without Alpha channels, the original Alpha value is 255.

(R/W)

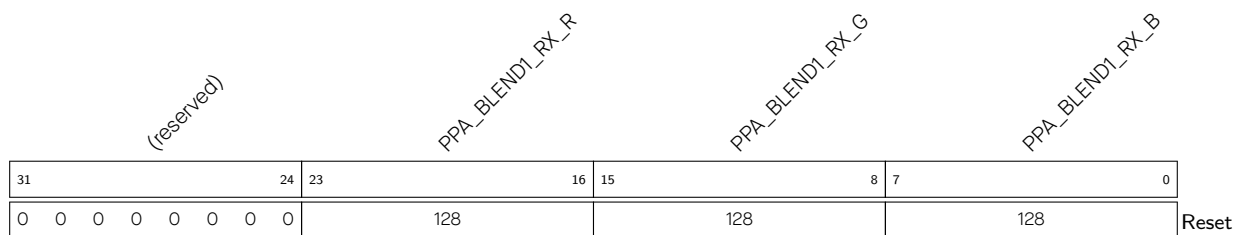
PPA_BLEND1_RX_ALPHA_INV Configures whether to invert the original Alpha value of the BLEND foreground layer data.

0: Not invert

1: Invert to 255 minus the original Alpha value

For data formats without Alpha channels, the original Alpha value is 255.

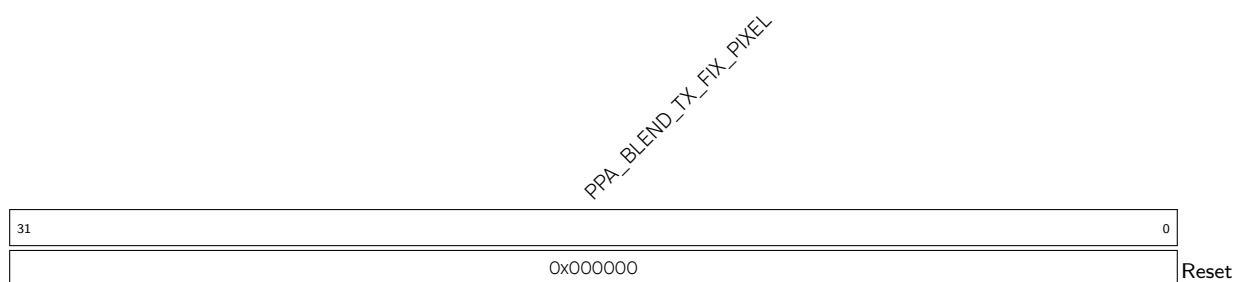
(R/W)

Register 3712. PPA_BLEND_RGB_REG (0x0048)

PPA_BLEND1_RX_B Configures the B channel value when the BLEND foreground layer is in A4/A8.
(R/W)

PPA_BLEND1_RX_G Configures the G channel value when the BLEND foreground layer is in A4/A8.
(R/W)

PPA_BLEND1_RX_R Configures the R channel value when the BLEND foreground layer is in A4/A8.
(R/W)

Register 3713. PPA_BLEND_FIX_PIXEL_REG (0x004C)

PPA_BLEND_TX_FIX_PIXEL Configures the pixel in the BLEND image filling mode.
[31:24]: A, [23:16]:R, [15:8]: G, [7:0]: B.
(R/W)

Register 37.14. PPA_CK_FG_LOW_REG (0x0050)

(reserved)								PPA_COLORKEY_FG_R_LOW								PPA_COLORKEY_FG_G_LOW								PPA_COLORKEY_FG_B_LOW								
31	24	23	16	15	8	7	0																									
0	0	0	0	0	0	0	0	0xff								0xff								0xff								Reset

PPA_COLORKEY_FG_B_LOW Configures the color-key lower threshold of the B channel value of the BLEND foreground layer. (R/W)

PPA_COLORKEY_FG_G_LOW Configures the color-key lower threshold of the G channel value of the BLEND foreground layer. (R/W)

PPA_COLORKEY_FG_R_LOW Configures the color-key lower threshold of the R channel value of the BLEND foreground layer. (R/W)

Register 37.15. PPA_CK_FG_HIGH_REG (0x0054)

(reserved)								PPA_COLORKEY_FG_R_HIGH								PPA_COLORKEY_FG_G_HIGH								PPA_COLORKEY_FG_B_HIGH								
3124								2316								158								70								
00000000								0x0								0x0								0x0								Reset

PPA_COLORKEY_FG_B_HIGH Configures the color-key higher threshold of the B channel value of the BLEND foreground layer. (R/W)

PPA_COLORKEY_FG_G_HIGH Configures the color-key higher threshold of the G channel value of the BLEND foreground layer. (R/W)

PPA_COLORKEY_FG_R_HIGH Configures the color-key higher threshold of the R channel value of the BLEND foreground layer. (R/W)

Register 37.16. PPA_CK_BG_LOW_REG (0x0058)

(reserved)								PPA_COLORKEY_BG_R_LOW								PPA_COLORKEY_BG_G_LOW								PPA_COLORKEY_BG_B_LOW								
3124								2316								158								70								
00000000								Oxff								Oxff								Oxff								Reset

PPA_COLORKEY_BG_B_LOW Configures the color-key lower threshold of the B channel value of the BLEND background layer. (R/W)

PPA_COLORKEY_BG_G_LOW Configures the color-key lower threshold of the G channel value of the BLEND background layer. (R/W)

PPA_COLORKEY_BG_R_LOW Configures the color-key lower threshold of the R channel value of the BLEND background layer. (R/W)

Register 37.17. PPA_CK_BG_HIGH_REG (0x005C)

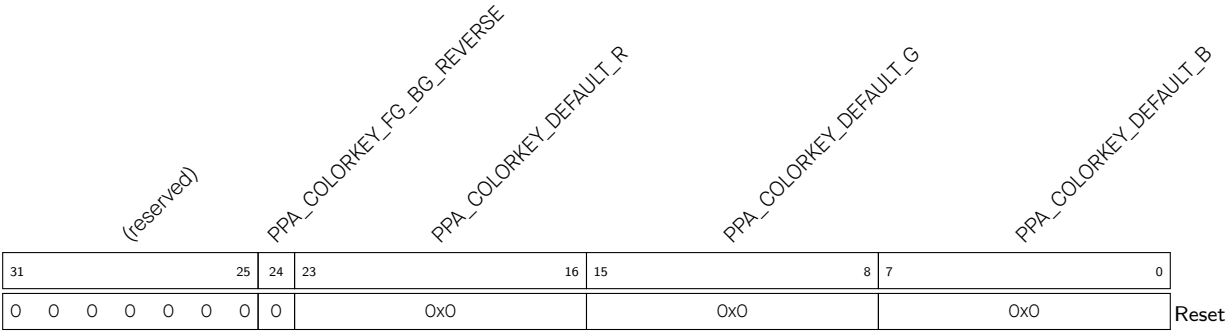
(reserved)								PPA_COLORKEY_BG_R_HIGH								PPA_COLORKEY_BG_G_HIGH								PPA_COLORKEY_BG_B_HIGH									
31	24	23	16	15	8	7	0																										
0	0	0	0	0	0	0	0	0	0x0								0x0								0x0								Reset

PPA_COLORKEY_BG_B_HIGH Configures the color-key higher threshold of the B channel value of the BLEND background layer. (R/W)

PPA_COLORKEY_BG_G_HIGH Configures the color-key higher threshold of the G channel value of the BLEND background layer. (R/W)

PPA_COLORKEY_BG_R_HIGH Configures the color-key higher threshold of the R channel value of the BLEND background layer. (R/W)

Register 37.18. PPA_CK_DEFAULT_REG (0x0060)



PPA_COLORKEY_DEFAULT_B Configures the default B channel value of BLEND color-key. (R/W)

PPA_COLORKEY_DEFAULT_G Configures the default G channel value of BLEND color-key. (R/W)

PPA_COLORKEY_DEFAULT_R Configures the default R channel value of BLEND color-key. (R/W)

PPA_COLORKEY_FG_BG_REVERSE Configures the workflow when when a pixel is within the back-ground color-key range but not within the foreground color-key range.

0: Output background pixel

1: Output foreground pixel

(R/W)

Register 37.19. PPA_SRM_SCAL_ROTATE_REG (0x0064)

(reserved) PPA_SRM_MIRROR_Y PPA_SRM_MIRROR_X PPA_SCAL_ROTATE_START PPA_SCAL_ROTATE_RST PPA_SRM_ROTATE_ANGLE PPA_SRM_SCAL_Y_FRAG PPA_SRM_SCAL_Y_INT PPA_SRM_SCAL_X_FRAG PPA_SRM_SCAL_X_INT																																																			
31	30	29	28	27	26	25	24	23	20										19	12										11	8										7	0									
0	0	0	0	0	0	0	0	0										1										0										1										Reset			

Reset

PPA_SRM_SCAL_X_INT Configures the integer part of the horizontal scaling factor for SRM. (R/W)

PPA_SRM_SCAL_X_FRAG Configures the fragment part of the horizontal scaling factor for SRM. (R/W)

PPA_SRM_SCAL_Y_INT Configures the integer part of the vertical scaling factor for SRM. (R/W)

PPA_SRM_SCAL_Y_FRAG Configures the fragment part of the vertical scaling factor for SRM. (R/W)

PPA_SRM_ROTATE_ANGLE Configures the counterclockwise rotation angle for the SRM.

- 0: 0 degree
 - 1: 90 degree
 - 2: 180 degree
 - 3: 270 degree
- (R/W)

PPA_SCAL_ROTATE_RST Configures whether to reset SRM.

- 0: Release reset
 - 1: Reset
- (R/W)

PPA_SCAL_ROTATE_START Write 1 to enable SRM. (WT)

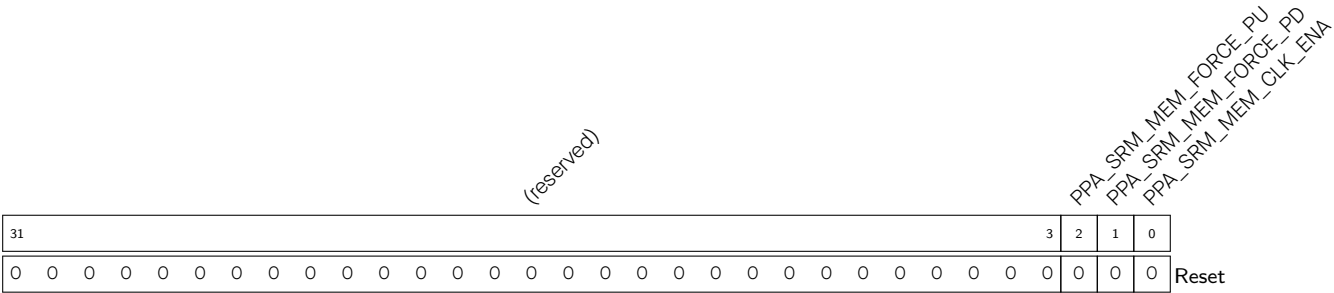
PPA_SRM_MIRROR_X Configures whether to enable the horizontal mirroring for SRM.

- 0: Disable
 - 1: Enable
- (R/W)

PPA_SRM_MIRROR_Y Configures whether to enable the vertical mirroring for SRM.

- 0: Disable
 - 1: Enable
- (R/W)

Register 37.20. PPA_SRM_MEM_PD_REG (0x0068)



PPA_SRM_MEM_CLK_ENA Configures whether to enable the force enable of the SRM MEM clock.

0: Disable
1: Enable
(R/W)

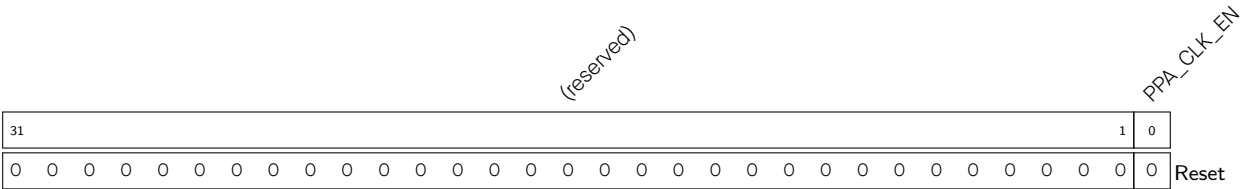
PPA_SRM_MEM_FORCE_PD Configures whether to enable the force power down of SRM MEM.

0: Disable
1: Enable
(R/W)

PPA_SRM_MEM_FORCE_PU Configures whether to enable the force power up of SRM MEM.

0: Disable
1: Enable
(R/W)

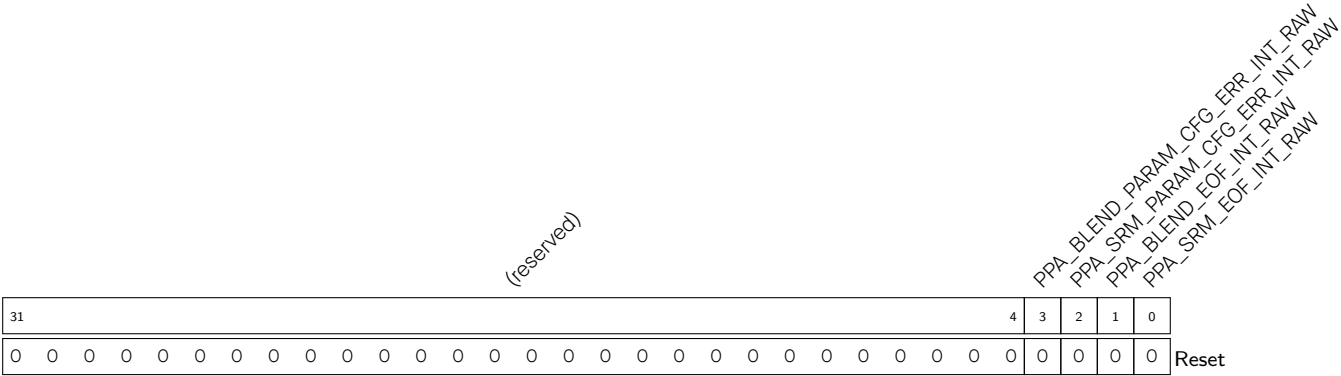
Register 37.21. PPA_REG_CONF_REG (0x006C)



PPA_CLK_EN Configures whether to keep the PPA register clock always on.

0: Clock only turns on when there's a register access
1: Clock always on
(R/W)

Register 37.22. PPA_INT_RAW_REG (0x0010)



PPA_SRM_EOF_INT_RAW The raw interrupt status of PPA_SRM_EOF_INT.
(R/WTC/SS)

PPA_BLEND_EOF_INT_RAW The raw interrupt status of PPA_BLEND_EOF_INT.
(R/WTC/SS)

PPA_SRM_PARAM_CFG_ERR_INT_RAW The raw interrupt status of PPA_SRM_PARAM_CFG_ERR_INT. Check [PPA_SRM_PARAM_ERR_ST_REG](#) to get the specific error.
(R/WTC/SS)

PPA_BLEND_PARAM_CFG_ERR_INT_RAW The raw interrupt status of PPA_BLEND_PARAM_CFG_ERR_INT. Check [PPA_BLEND_ST_REG](#) to get the specific error.
(R/WTC/SS)

Register 37.23. PPA_INT_ST_REG (0x0014)

(reserved)																																PPA_BLEND_PARAM_CFG_ERR_INT_ST PPA_SRM_PARAM_CFG_ERR_INT_ST PPA_BLEND_EOF_INT_ST PPA_SRM_EOF_INT_ST																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
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PPA_SRM_EOF_INT_ST The masked interrupt status of PPA_SRM_EOF_INT. (RO)

PPA_BLEND_EOF_INT_ST The masked interrupt status of PPA_BLEND_EOF_INT. (RO)

PPA_SRM_PARAM_CFG_ERR_INT_ST The masked interrupt status of PPA_SRM_PARAM_CFG_ERR_INT. (RO)

PPA_BLEND_PARAM_CFG_ERR_INT_ST The masked interrupt status of PPA_BLEND_PARAM_CFG_ERR_INT. (RO)

Register 37.24. PPA_INT_ENA_REG (0x0018)

(reserved)																												PPA_BLEND_PARAM_CFG_ERR_INT_ENA PPA_SRM_PARAM_CFG_ERR_INT_ENA PPA_BLEND_EOF_INT_ENA PPA_SRM_EOF_INT_ENA																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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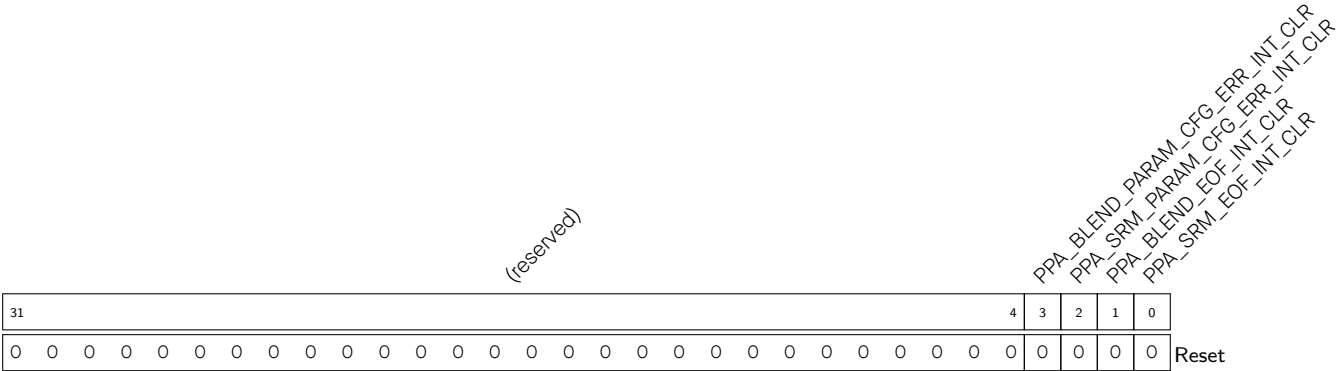
PPA_SRM_EOF_INT_ENA Write 1 to enable PPA_SRM_EOF_INT. (R/W)

PPA_BLEND_EOF_INT_ENA Write 1 to enable PPA_BLEND_EOF_INT. (R/W)

PPA_SRM_PARAM_CFG_ERR_INT_ENA Write 1 to enable PPA_SRM_PARAM_CFG_ERR_INT. (R/W)

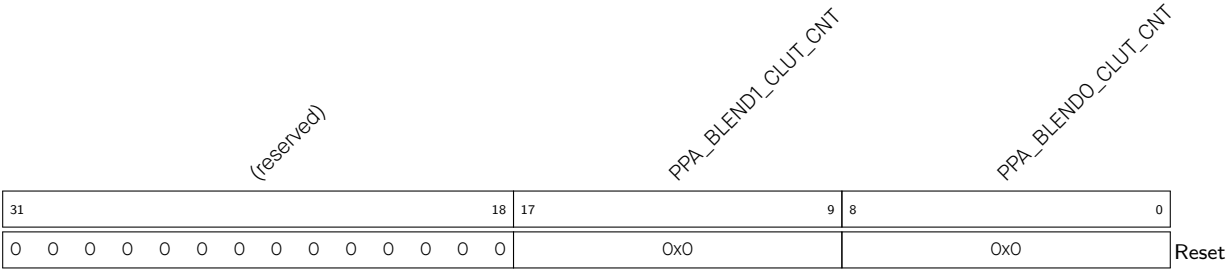
PPA_BLEND_PARAM_CFG_ERR_INT_ENA Write 1 to enable PPA_BLEND_PARAM_CFG_ERR_INT. (R/W)

Register 37.25. PPA_INT_CLR_REG (0x001C)



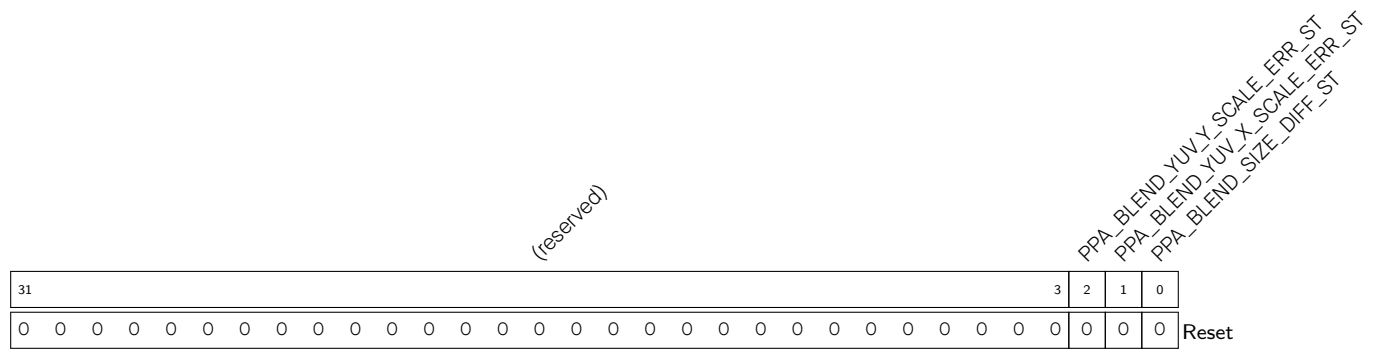
- PPA_SRM_EOF_INT_CLR Write 1 to clear PPA_SRM_EOF_INT. (WT)
- PPA_BLEND_EOF_INT_CLR Write 1 to clear PPA_BLEND_EOF_INT. (WT)
- PPA_SRM_PARAM_CFG_ERR_INT_CLR Write 1 to clear PPA_SRM_PARAM_CFG_ERR_INT. (WT)
- PPA_BLEND_PARAM_CFG_ERR_INT_CLR □ 1 □□ PPA_BLEND_PARAM_CFG_ERR_INT□ (WT)

Register 37.26. PPA_CLUT_CNT_REG (0x0070)



- PPA_BLEND0_CLUT_CNT Represents the current write address of the BLEND background layer CLUT in FIFO mode. (RO)
- PPA_BLEND1_CLUT_CNT Represents the current write address of the BLEND foreground layer CLUT in FIFO mode. (RO)

Register 37.27. PPA_BLEND_ST_REG (0x0074)



PPA_BLEND_SIZE_DIFF_ST	Represents whether the size of the BLEND background and foreground images are consistent.
-------------------------------	---

0: Consistent

1: Different

(RO)

PPA_BLEND_YUV_X_SCALE_ERR_ST	Represents the image width (X) is odd for YUV422 or YUV420 formats. (RO)
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PPA_BLEND_YUV_Y_SCALE_ERR_ST Represents the image height (Y) is odd for the YUV420 format.
(RO)

Register 37.28. PPA_SRM_PARAM_ERR_ST_REG (0x0078)

(reserved)																PPA_Y_YUV420_TX_SCALE_ERR_ST PPA_X_YUV420_TX_SCALE_ERR_ST PPA_Y_YUV420_RX_SCALE_ERR_ST PPA_X_YUV420_RX_SCALE_ERR_ST PPA_XDST_LEN_TOO_LARGE_ERR_ST PPA_XDST_LEN_TOO_SAMLL_ERR_ST PPA_RX_DSCR_HB_ERR_ST PPA_X_RX_SCAL_EQUAL_O_ERR_ST PPA_YDST_LEN_TOO_LARGE_ERR_ST PPA_YDST_LEN_TOO_SAMLL_ERR_ST PPA_Y_RX_DSCR_VB_ERR_ST PPA_TX_DSCR_HB_ERR_ST PPA_TX_DSCR_VB_ERR_ST															
31																14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset		

PPA_TX_DSCR_VB_ERR_ST Represents whether the vertical size of the SRM output image block plus the vertical offset Y configured in the output list exceeds the vertical size of the entire image configured in the output list, VA.

0: Not exceeded
1: Exceeded
(RO)

PPA_TX_DSCR_HB_ERR_ST Represents whether the horizontal size of the SRM output image block plus the vertical offset X configured in the output list exceeds the horizontal size of the entire image configured in the output list, HA.

0: Not exceeded
1: Exceeded
(RO)

PPA_Y_RX_SCAL_EQUAL_O_ERR_ST Represents whether the vertical direction scaling factor of the SRM input image block is 0.

0: Not 0
1: 0
(RO)

PPA_RX_DSCR_VB_ERR_ST Represents whether the sum of the vertical direction image block size VB configured in the SRM input list and the vertical offset Y exceeds the entire image's vertical size VA.

0: Not exceeded
1: Exceeded
(RO)

PPA_YDST_LEN_TOO_SAMLL_ERR_ST Represents whether the vertical size of the image block after scaling in the SRM is 0.

0: Not 0
1: 0
(RO)

Continued on the next page...

Register 37.28. PPA_SRM_PARAM_ERR_ST_REG (0x0078)

Continued from the previous page...

PPA_YDST_LEN_TOO_LARGE_ERR_ST Represents whether the vertical size of the image block after scaling in the SRM is larger than 8191.

0: Within 8191

1: Larger than 8191

(RO)

PPA_X_RX_SCAL_EQUAL_O_ERR_ST Represents whether the horizontal direction scaling factor of the SRM input image block is 0.

0: Not 0

1: 0

(RO)

PPA_RX_DSCR_HB_ERR_ST Represents whether the sum of the horizontal direction image block size HB configured in the SRM input list and the horizontal offset X exceeds the entire image's horizontal size HA.

0: Not exceeded

1: Exceeded

(RO)

PPA_XDST_LEN_TOO_SAMLL_ERR_ST Represents whether the horizontal size of the image block after scaling in the SRM is 0.

0: Not 0

1: 0

(RO)

PPA_XDST_LEN_TOO_LARGE_ERR_ST Represents whether the horizontal size of the image block after scaling in the SRM is larger than 8191.

0: Within 8191

1: Larger than 8191

(RO)

PPA_X_YUV420_RX_SCALE_ERR_ST Represents whether the parameters HA/HB/X of the SRM input list are odd when the input format is YUV420.

0: Even

1: Odd

(RO)

PPA_Y_YUV420_RX_SCALE_ERR_ST Represents whether the parameters VA/VB/Y of the SRM input list are odd when the input format is YUV420.

0: Even

1: Odd

(RO)

Continued on the next page...

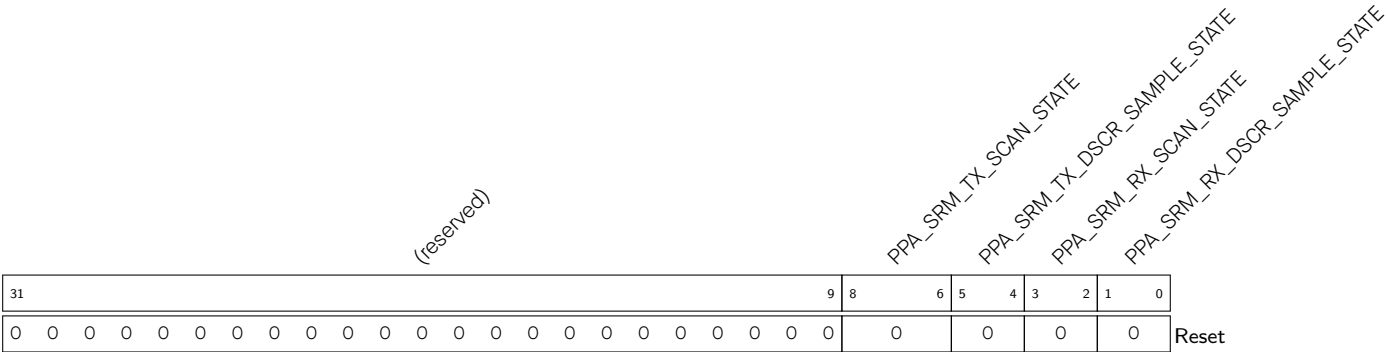
Register 37.28. PPA_SRM_PARAM_ERR_ST_REG (0x0078)

Continued from the previous page...

PPA_X_YUV420_TX_SCALE_ERR_ST Represents whether the parameters HA/HB/X of the SRM output list are odd when the output format is YUV420.
0: Even
1: Odd
(RO)

PPA_Y_YUV420_TX_SCALE_ERR_ST Represents whether the parameters VA/VB/Y of the SRM output list are odd when the output format is YUV420.
0: Even
1: Odd
(RO)

Register 37.29. PPA_SRM_STATUS_REG (0x007C)



PPA_SRM_RX_DSCR_SAMPLE_STATE Represents the SRM RX descriptor state. (RO)

PPA_SRM_RX_SCAN_STATE Represents the SRM RX pixel scan state. (RO)

PPA_SRM_TX_DSCR_SAMPLE_STATE Represents the SRM TX descriptor state. (RO)

PPA_SRM_TX_SCAN_STATE Represents the SRM TX pixel scan state. (RO)

Register 37.30. PPA_DEBUG_CTRL0_REG (0x0090)

(reserved)																															PPA_DBG_REPLACE_SEL		
31																															3	2	0
0 0																															0x0		Reset

PPA_DBG_REPLACE_SEL Configures where the replacement pixel is inserted.

- 0: No replacement
 - 1: SRM RX input
 - 2: SRM interpolation output
 - 3: SRM TX output
 - 4: BLEND foreground input
 - 5: BLEND background input
 - 6: BLEND output
- (R/W)

Register 37.31. PPA_DEBUG_CTRL1_REG (0x0094)

PPA_DBG_REPLACE_DATA																															
31																															0
0x000000																															
Reset																															

PPA_DBG_REPLACE_DATA Configures the replacement pixel value. (R/W)

Register 37.32. PPA_RGB2GRAY_REG (0x0098)

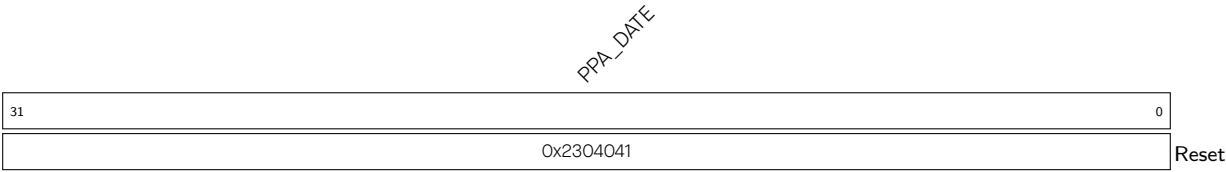
(reserved)								PPA_RGB2GRAY_R								PPA_RGB2GRAY_G								PPA_RGB2GRAY_B																																																																							
31								24								23								16								15								8								7								0																																							
0								0								0								0								0								0								0								0								85								86								85								Reset							

PPA_RGB2GRAY_B Configures the coefficient for B in RGB to GRAY conversion. (R/W)

PPA_RGB2GRAY_G Configures the coefficient for G in RGB to GRAY conversion. (R/W)

PPA_RGB2GRAY_R Configures the coefficient for R in RGB to GRAY conversion. (R/W)

Register 37.33. PPA_DATE_REG (0x0100)



PPA_DATE Version control register. (R/W)